A General Survey on Attention Mechanisms in Deep Learning

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Abstract—Attention is an important mechanism that can be employed for a variety of deep learning models across many different domains and tasks. This survey provides an overview of the most important attention mechanisms proposed in the literature. The various attention mechanisms are explained by means of a framework consisting of a general attention model, uniform notation, and a comprehensive taxonomy of attention mechanisms. Furthermore, the various measures for evaluating attention models are reviewed, and methods to characterize the structure of attention models based on the proposed framework are discussed. Last, future work in the field of attention models is considered.

Index Terms—Attention models, deep learning, introductory and survey, neural nets, supervised learning

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1 Introduction

The idea of mimicking human attention first arose in the field of computer vision [1], [2] in an attempt to reduce the computational complexity of image processing while improving performance by introducing a model that would only focus on specific regions of images instead of the entire picture. Although, the true starting point of the attention mechanisms we know today is often attributed to originate in the field of natural language processing [3]. Bahdanau et al. [3] implement attention in a machine translation model to address certain issues with the structure of recurrent neural networks. After Bahdanau et al. [3] emphasized the advantages of attention, the attention techniques were refined [4] and quickly became popular for a variety of tasks, such as text classification [5], [6], image captioning [7], [8], sentiment analysis [6], [9], and speech recognition [10], [11], [12].

Attention has become a popular technique in deep learning for several reasons. Firstly, models that incorporate attention mechanisms attain state-of-the-art results for all of the previously mentioned tasks, and many others. Furthermore, most attention mechanisms can be trained jointly with a base model, such as a recurrent neural network or a convolutional neural network using regular backpropagation [3]. Additionally, attention introduces a certain type of interpretation into neural network models [8] that are generally known to be highly complicated to interpret. Moreover, the popularity of attention mechanisms was additionally boosted after the introduction of the Transformer model [13] that further proved how effective attention can be. Attention was originally introduced as an extension to recurrent neural networks [14]. However, the Transformer model proposed in [13] poses a major development in attention research as it demonstrates that the attention mechanism is sufficient to build a state-of-the-art model. This means that disadvantages, such as the fact that recurrent neural networks are particularly difficult to parallelize, can be circumvented. As was the case for the introduction of the original attention mechanism [3], the Transformer model was created for machine translation, but was quickly adopted to be used for other tasks, such as image processing [15], video processing [16], and recommender systems [17].

The purpose of this survey is to explain the general form of attention, and provide a comprehensive overview of attention techniques in deep learning. Other surveys have already been published on the subject of attention models. For example, in [18], a survey is presented on attention in computer vision, [19] provides an overview of attention in graph models, and [20], [21], [22] are all surveys on attention in natural language processing. This paper partly builds on the information presented in the previously mentioned surveys. Yet, we provide our own significant contributions. The main difference between this survey and the previously mentioned ones is that the other surveys generally focus on attention models within a certain domain. This survey, however, provides a cross-domain overview of attention techniques. We discuss the attention techniques in a general way, allowing them to be understood and applied in a variety of domains. Furthermore, we found the taxonomies presented in previous surveys to be lacking the depth and structure needed to properly distinguish the various attention mechanisms. Additionally, certain significant attention techniques have not yet been properly discussed in previous surveys, while other presented attention mechanisms seem to be lacking either technical details or intuitive explanations. Therefore, in this paper, we present important attention techniques by means of a single framework using a uniform notation, a combination of both technical and intuitive explanations for each presented attention technique, and a comprehensive taxonomy of attention mechanisms.

The structure of this paper is as follows. Section 2 introduces a general attention model that provides the reader with a basic understanding of the properties of attention and how it can be applied. One of the main contributions of this paper is the taxonomy of attention techniques presented in Section 3. In this section, attention mechanisms are explained and categorized according to the presented

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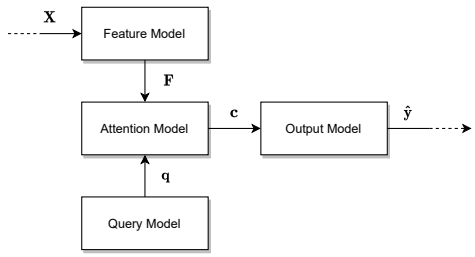

Fig. 1. An illustration of the general structure of the task model.

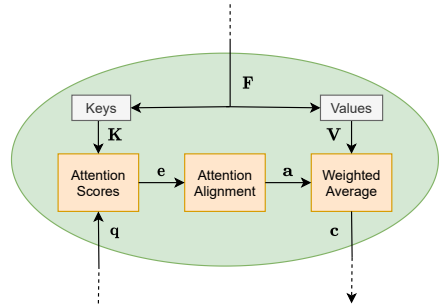

Fig. 2. The inner mechanisms of the general attention module.

taxonomy. Section 4 provides an overview of performance measures and methods for evaluating attention models. Furthermore, the taxonomy is used to evaluate the structure of various attention models. Lastly, in Section 5, we give our conclusions and suggestions for further research.

2 General Attention Model

This section presents a general form of attention with corresponding notation. The notation introduced here is based on the notation that was introduced in [23] and popularized in [13]. The framework presented in this section is used throughout the rest of this paper.

To implement a general attention model, it is necessary to first describe the general characteristics of a model that can employ attention. First of all, we will refer to the complete model as the **task model**, of which the structure is presented in Fig. 1. This model simply takes an input, carries out the specified task, and produces the desired output. For example, the task model can be a language model that takes as input a piece of text, and produces as output a summary of the contents, a classification of the sentiment, or the text translated word for word to another language. Alternatively, the task model can take an image, and produce a caption or segmentation for that image. The task model consists of four submodels: the feature model, the query model, the attention model, and the output model. In Subsection 2.1, the feature model and query model are discussed, which are used to prepare the input for the attention calculation. In Subsection 2.2, the attention model and output model are discussed, which are concerned with producing the output.

2.1 Attention Input

Suppose the task model takes as input the matrix $\boldsymbol{X} \in \mathbb{R}^{d_x \times n_x}$, where d_x represents the size of the input vectors and n_x represents the amount of input vectors. The columns in this matrix can represent the words in a sentence, the pixels in an image, the characteristics of an acoustic sequence, or any other collection of inputs. The **feature model** is then employed to extract the n_f feature vectors $\boldsymbol{f}_1, \ldots, \boldsymbol{f}_{n_f} \in \mathbb{R}^{d_f}$ from $\boldsymbol{X}$, where d_f represents the size of the feature vectors. The feature model can be a recurrent neural network (RNN), a convolutional neural network (CNN), a simple embedding layer, a linear transformation of the original data, or no transformation at all. Essentially, the feature model consists of all the steps that transform the original input $\boldsymbol{X}$ into the feature vectors $\boldsymbol{f}_1, \ldots, \boldsymbol{f}_{n_f}$ that the attention model will attend to.

To determine which vectors to attend to, the attention model requires the query $\boldsymbol{q} \in \mathbb{R}^{d_q}$, where d_q indicates the size of the query vector. This query is extracted by the **query model**, and is generally designed based on the type of output that is desired of the model. A query tells the attention model which feature vectors to attend to. It can be interpreted literally as a query, or a question. For example, for the task of image captioning, suppose that one uses a decoder RNN model to produce the output caption based on feature vectors obtained from the image by a CNN. At each prediction step, the hidden state of the RNN model can be used as a query to attend to the CNN feature vectors. In each step, the query is a question in the sense that it asks for the necessary information from the feature vectors based on the current prediction context.

2.2 Attention Output

The feature vectors and query are used as input for the **attention model**. This model consists of a single, or a collection of **general attention modules**. An overview of a general attention module is presented in Fig. 2. The input of the general attention module is the query $\boldsymbol{q} \in \mathbb{R}^{d_q}$, and the matrix of feature vectors $\boldsymbol{F} = [\boldsymbol{f}_1, \ldots, \boldsymbol{f}_{n_f}] \in \mathbb{R}^{d_f \times n_f}$. Two separate matrices are extracted from the matrix $\boldsymbol{F}$: the keys matrix $\boldsymbol{K} = [\boldsymbol{k}_1, \ldots, \boldsymbol{k}_{n_f}] \in \mathbb{R}^{d_k \times n_f}$, and the values matrix $\boldsymbol{V} = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{n_f}] \in \mathbb{R}^{d_v \times n_f}$, where d_k and d_v indicate, respectively, the dimensions of the key vectors (columns of $\boldsymbol{K}$) and value vectors (columns of $\boldsymbol{V}$). The general way of obtaining these matrices is through a linear transformation of $\boldsymbol{F}$ using the weight matrices $\boldsymbol{W}_K \in \mathbb{R}^{d_k \times d_f}$ and $\boldsymbol{W}_V \in \mathbb{R}^{d_v \times d_f}$, for $\boldsymbol{K}$ and $\boldsymbol{V}$, respectively. The calculations of $\boldsymbol{K}$ and $\boldsymbol{V}$ are presented in (1). Both weight matrices can be learned during training or predefined by the researcher. For example, one can choose to define both $\boldsymbol{W}_K$ and $\boldsymbol{W}_V$ as equal to the identity matrix to retain the original feature vectors. Other ways of defining the keys and the values are also possible, such as using completely separate inputs for the keys and values. The only constraint to be obeyed is that the number of columns in $\boldsymbol{K}$ and $\boldsymbol{V}$ remains the same.

$$\underset{d_k \times n_f}{\boldsymbol{K}} = \underset{d_k \times d_f}{\boldsymbol{W}_K} \times \underset{d_f \times n_f}{\boldsymbol{F}}, \qquad \underset{d_v \times n_f}{\boldsymbol{V}} = \underset{d_v \times d_f}{\boldsymbol{W}_V} \times \underset{d_f \times n_f}{\boldsymbol{F}}. \tag{1}$$

The goal of the attention module is to produce a weighted average of the value vectors in $\boldsymbol{V}$. The weights used to produce this output are obtained via an attention scoring and alignment step. The query $\boldsymbol{q}$ and the keys matrix $\boldsymbol{K}$ are used to calculate the vector of attention scores

$\boldsymbol{e} = [e_1, \ldots, e_{n_f}] \in \mathbb{R}^{n_f}$. This is done via the score function score(), as illustrated in (2).

$$\underset{1\times 1}{e_l} = \text{score}(\underset{d_q\times 1}{\boldsymbol{q}}, \underset{d_k\times 1}{\boldsymbol{k}_l}). \tag{2}$$

As discussed before, the query symbolizes a request for information. The attention score e_l represents how important the information contained in the key vector $\boldsymbol{k}_l$ is according to the query. If the dimensions of the query and key vectors are the same, an example of a score function would be to take the dot-product of the vectors. The different types of score functions are further discussed in Section 3.2.1.

Next, the attention scores are processed further through an alignment layer. The attention scores can generally have a wide range outside of $[0, 1]$. However, since the goal is to produce a weighted average, the scores are redistributed via an alignment function align() as defined in (3).

$$\underset{1\times 1}{a_l} = \text{align}(\underset{1\times 1}{e_l}; \underset{n_f\times 1}{\boldsymbol{e}}), \tag{3}$$

where $a_l \in \mathbb{R}^1$ is the attention weight corresponding to the lth value vector. One example of an alignment function would be to use a softmax function, but the various other alignment types are discussed in Section 3.2.2. The attention weights provide a rather intuitive interpretation for the attention module. Each weight is a direct indication of how important each feature vector is relative to the others for this particular problem. This can provide us with a more in-depth understanding of the model behaviour, and the relations between inputs and outputs. The vector of attention weights $\boldsymbol{a} = [a_1, \ldots, a_{n_f}] \in \mathbb{R}^{n_f}$ is used to produce the context vector $\boldsymbol{c} \in \mathbb{R}^{d_v}$ by calculating a weighted average of the columns of the values matrix $\boldsymbol{V}$, as shown in (4).

$$\underset{d_v\times 1}{\boldsymbol{c}} = \sum_{l=1}^{n_f} \underset{1\times 1}{a_l} \times \underset{d_v\times 1}{\boldsymbol{v}_l}. \tag{4}$$

As illustrated in Fig. 1, the context vector is then used in the **output model** to create the output $\hat{\boldsymbol{y}}$. This output model translates the context vector into an output prediction. For example, it could be a simple softmax layer that takes as input the context vector $\boldsymbol{c}$, as shown in (5).

$$\underset{d_y\times 1}{\hat{\boldsymbol{y}}} = \text{softmax}(\underset{d_y\times d_v}{\boldsymbol{W}_c} \times \underset{d_v\times 1}{\boldsymbol{c}} + \underset{d_y\times 1}{\boldsymbol{b}_c}), \tag{5}$$

where d_y is the number of output choices or classes, and $\boldsymbol{W}_c \in \mathbb{R}^{d_y\times d_v}$ and $\boldsymbol{b}_c \in \mathbb{R}^{d_y}$ are trainable weights.

2.3 Attention Applications

Attention is a rather general mechanism that can be used in a wide variety of problem domains. Consider the task of machine translation using an RNN model. Also, consider the problem of image classification using a basic CNN model. While an RNN produces a sequence of hidden state vectors, a CNN creates feature maps, where each region in the image is represented by a feature vector. The RNN hidden states are organized sequentially, while the CNN feature maps are organized spatially. Yet, attention can still be applied in both situations, since the attention mechanism does not inherently depend on the organization of the feature vectors. This characteristic makes attention easy to implement in a wide variety of models in different domains.

Another domain where attention can be applied is audio processing [24], [25]. Acoustic sequences can be represented by a sequence of feature vectors that relate to certain time periods of the audio sample. These vectors could simply be the raw input audio, or they can be extracted via, for example, an RNN or CNN. Video processing is another domain where attention can be applied intuitively [26], [27]. Video data consists of sequences of images, so attention can be applied to the individual images, as well as the entire sequence. Recommender systems often incorporate a user's interaction history to produce recommendations. Feature vectors can be extracted based on, for example, the id's or other characteristics of the products the user interacted with, and attention can be applied to them [28]. Attention can generally also be applied to many problems that use a time series as input, be it medical [29], financial [30], or anything else, as long as feature vectors can be extracted.

The fact that attention does not rely on the organization of the feature vectors allows it to be applied to various problems that each use data with different structures, as illustrated by the previous domain examples. Yet, this can be taken even further by applying attention to data where there is irregular structure. For example, protein structures, city traffic flows, and communication networks cannot always be represented using neatly structured organizations, such as sequences, like time series, or grids, like images. In such cases, the different aspects of the data are often represented as nodes in a graph. These nodes can be represented by feature vectors, meaning that attention can be applied in domains that use graph-structured data as well [19], [31].

In general, attention can be applied to any problem for which a set of feature vectors can be defined or extracted. As such, the general attention model presented in Fig. 2 is applicable to a wide range of domains. The problem, however, is that there is a large variety of different applications and extensions of the general attention module. As such, in Section 3, a comprehensive overview is provided of a collection of different attention mechanisms.

3 Attention Taxonomy

There are many different types of attention mechanisms and extensions, and a model can use different combinations of these attention techniques. As such, we propose a taxonomy that can be used to classify different types of attention mechanisms. Fig. 3 provides a visual overview of the different categories and subcategories that the attention mechanisms can be organized in. The three major categories are based on whether an attention technique is designed to handle specific types of feature vectors (feature-related), specific types of model queries (query-related), or whether it is simply a general mechanism that is related to neither the feature model, nor the query model (general). Further explanations of these categories and their subcategories are provided in the following subsections. Each mechanism discussed in this section is either a modification to the existing inner mechanisms of the general attention module presented in Section 2, or an extension of it.

The presented taxonomy can also be used to analyze the architecture of attention models. Namely, the major categories and their subcategories can be interpreted as